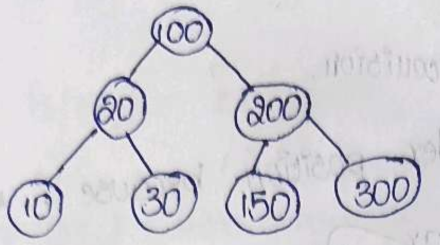
ACTIVITY-6

CO3-AT3- TREE TRAVERSAL

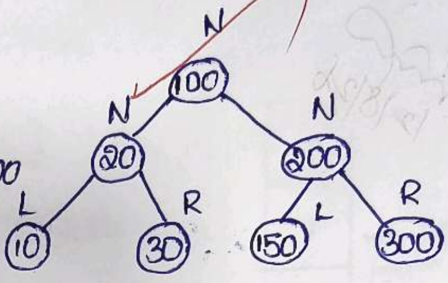
1.



TREE TRANVERSAL

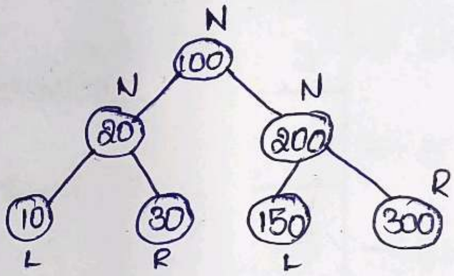
i, INORDER: (LNR)

$\underset{L}{\underline{10\ 20\ 30}}$ $\underset{100}{\underline{N}}$ $\underset{R.}{\underline{150\ 200\ 300}}$



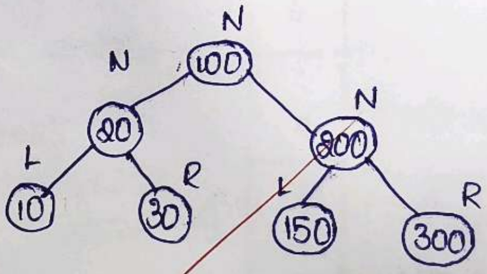
ii, PREORDER: (NLR)

$\underset{N}{\underline{100}}$ $\underset{L}{\underline{20\ 10\ 30}}$ $\underset{R.}{\underline{200\ 150\ 300}}$



iii, POSTORDER (LRN)

$\underset{L}{\underline{10\ 30\ 20}}$ $\underset{R}{\underline{150\ 300\ 200}}$ $\underset{N}{\underline{100}}$

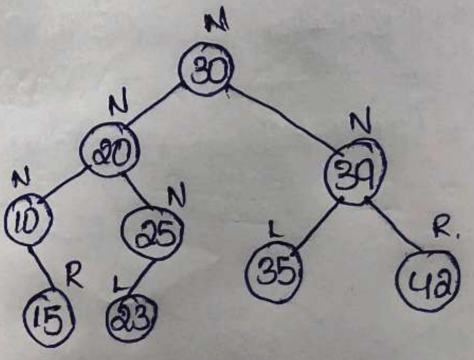


(2) TREE TRAVERSAL:

i INORDER: (LNR)

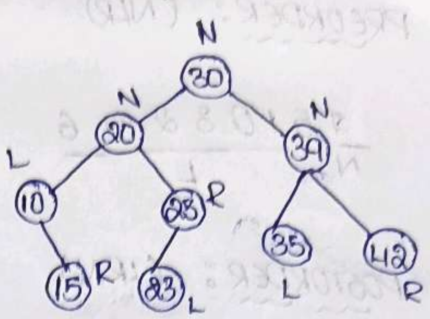
$\underset{L}{\underline{10\ 15\ 20\ 23\ 25}}$ $\underset{N}{\underline{30}}$

$\underset{R}{\underline{35\ 39\ 42}}$



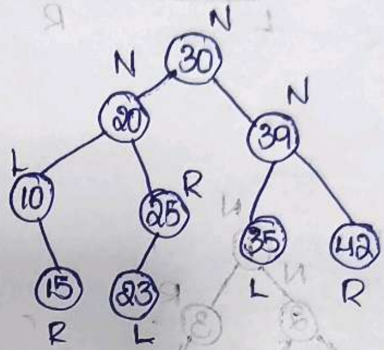
ii, PREORDER: (NLR)

$\underset{N}{30}$ $\underset{L}{20\ 10\ 15\ 25\ 23}$ $\underset{R}{39\ 35\ 42}$



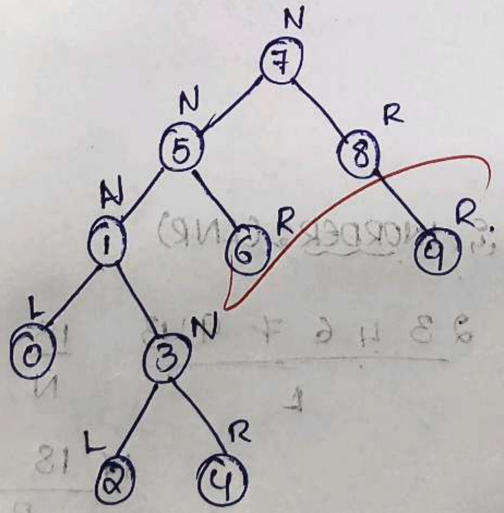
iii, POSTORDER: (LRN)

$\underset{L}{15\ 10\ 23\ 25\ 20}$ $\underset{R}{42\ 39\ 35}$ $\underset{N}{30}$



(3) Assume the digits 7,5,1,8,3,6,0,4,2 are added in that sequence into a binary search tree that is initialy empty. The binary search tree follows standard natural number ordering. what is the Resulting tree tranvers sequal. ?

A.



i, INORDER: (LNR

$\underset{L}{0\ 1\ 2\ 3\ 4\ 5\ 6}$ $\underset{N}{7}$ $\underset{R}{8\ 9}$

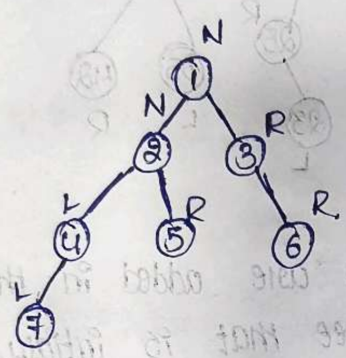
ii, PREORDER : (NLR)

$\underbrace{7}_{N}\ \underbrace{5\,1\,0\,3\,2\,4\,6}_{L}\ ,\ \underbrace{8\,9}_{R}$

iii, POSTORDER : (LRN)

$\underbrace{0\,2\,4\,3\,1\,6\,5}_{L}\quad \underbrace{9\,8}_{R}\quad \underbrace{7}_{N}$

(5)



i, INORDER : (LNR)

$\underbrace{7\,4\,2\,5}_{L}\quad \underbrace{1}_{N}\quad \underbrace{36}_{R.}$

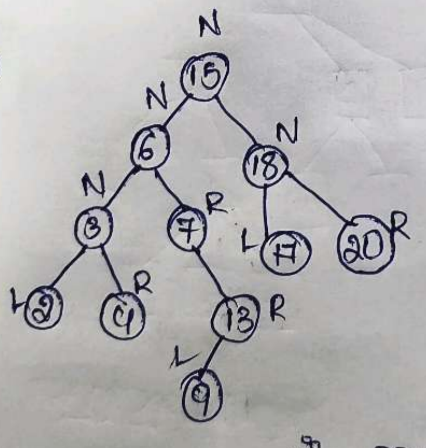
ii, PRE ORDER : (NLR)

$\underbrace{1}_{N}\ \underbrace{2\,4\,7\,5}_{L}\ \underbrace{3\,6}_{R.}$

iii, POSTORDER : (LRN)

$\underbrace{7\,4\,5\,2}_{L}\quad \underbrace{6\,3}_{R}\quad \underbrace{1}_{N}$

(6)



i, INORDER : (LNR)

$\underbrace{2\ 3\ 4\ 6\ 7\ 9\ 13}_{L}\quad \underbrace{15}_{N}$

$\underbrace{17\ 18\ 20}_{R.}$

ii, PREORDER : (NLR)

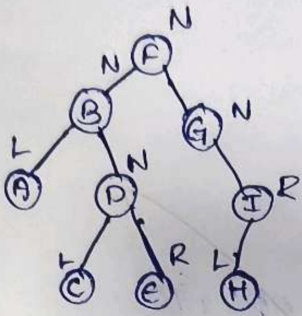
$\underbrace{15}_{N}\ \underbrace{6\ 3\ 2\ 4\ 7\ 13\ 9}_{L}\ \underbrace{18\ 17\ 20}_{R}$

iii) POSTORDER : (LRN)

$\frac{2\ 4\ 3\ 9\ 13\ 7\ 6}{L}$ $\frac{17\ 20\ 13}{R}$ $\frac{15}{N}$

(7) What is the sequence of nodes when applying in order tranvel on BT given below?

A.



INORDER : (LNR)

$\frac{A\ B\ C\ D\ E}{L}$ $\frac{F}{N}$ $\frac{G\ H\ I}{R.}$

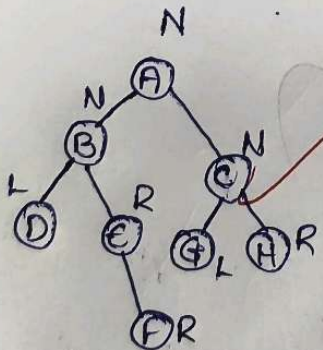
(8) consider the following inorder & preorder tranvel of

INORDER : DBEFAGHC

PREORDER : ABDEFCGH

WHAT is the postorder Tranversal same tree.

A.

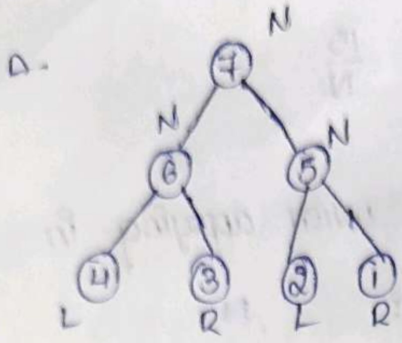


POSTORDER : (LRN)

$\frac{DFEB}{L}$ $\frac{GHC}{R}$ $\frac{A}{N}$

(9) Find INORDER, PREORDER & POSTORDER

A.



i) INORDER : (LNR)

$\underbrace{4\ 6\ 3}_{L}\quad \underbrace{7}_{N}\quad \underbrace{2\ 5\ 1}_{R}$

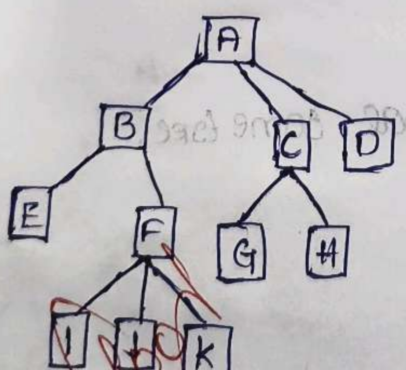
ii) PREORDER : (NLR)

$\underbrace{7}_{N}\quad \underbrace{6\ 4\ 3}_{L}\quad \underbrace{5\ 2\ 1}_{R}$

iii) POSTORDER : (LRN)

$\underbrace{4\ 3\ 6}_{L}\quad \underbrace{2\ 1\ 5}_{R}\quad \underbrace{7}_{N}$

(10) TREE TRAVERSAL:



PREORDER : (NLR)

$\underbrace{A}_{N}\ \underbrace{B\,E\,F\,I\,J\,K}_{L}\ \underbrace{C\,G\,H\,D}_{R}$

POSTORDER : (LRN)

$\underbrace{E\,I\,J\,K\,F\,B}_{L}\ \underbrace{G\,H\,C\,D}_{R}\ \underbrace{A}_{N}$